

Waste Collection & Recycling Management Portal

Problem Statement

Urban areas generate a significant amount of waste daily, including household, commercial, and industrial waste. Managing this waste efficiently is a major challenge for municipalities and private waste management agencies. In many cities, waste collection and recycling processes are still partially manual, fragmented, or poorly coordinated.

Citizens often lack awareness or tools to properly dispose of waste, while authorities struggle with tracking collection activities, monitoring staff performance, and ensuring proper recycling practices.

Current Challenges

Lack of Centralized System

Waste collection data is often maintained across different systems, leading to inefficiency, duplication, and data loss.

Inefficient Waste Collection Scheduling

Manual planning of collection routes leads to missed pickups, overlapping routes, and increased fuel costs.

Limited Citizen Engagement

Residents do not have a proper platform to report uncollected waste, request pickups, or track complaint status.

Poor Monitoring of Waste Segregation

There is minimal tracking of whether waste is properly segregated (dry, wet, recyclable), reducing recycling efficiency.

Delay in Issue Resolution

Complaints regarding missed collections or overflowing bins take time to resolve due to a lack of structured workflow.

Lack of Transparency

Citizens and administrators cannot easily track collection status, recycling rates, or operational performance.

Inefficient Data Analysis

Authorities struggle to generate insights such as waste volume trends, recycling efficiency, and area-wise performance.

Security and Data Integrity Issues

Existing systems may lack proper user authentication, making data vulnerable to unauthorized access.

Need for the Proposed System

To address these issues, there is a need for a **web-based and/or mobile-based Waste Collection & Recycling Management Portal** that:

- Centralizes all waste-related data
 - Automates scheduling and tracking
 - Increases citizen participation
 - Improves recycling efficiency
 - Ensures transparency and accountability
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Proposed Solution

The **Waste Collection & Recycling Management Portal** will be a role-based system with the following core users:

Citizens

- Register and log in
- Request waste pickup
- Report uncollected waste or issues
- Track complaint/request status
- Get awareness notifications on waste segregation

Collection Staff

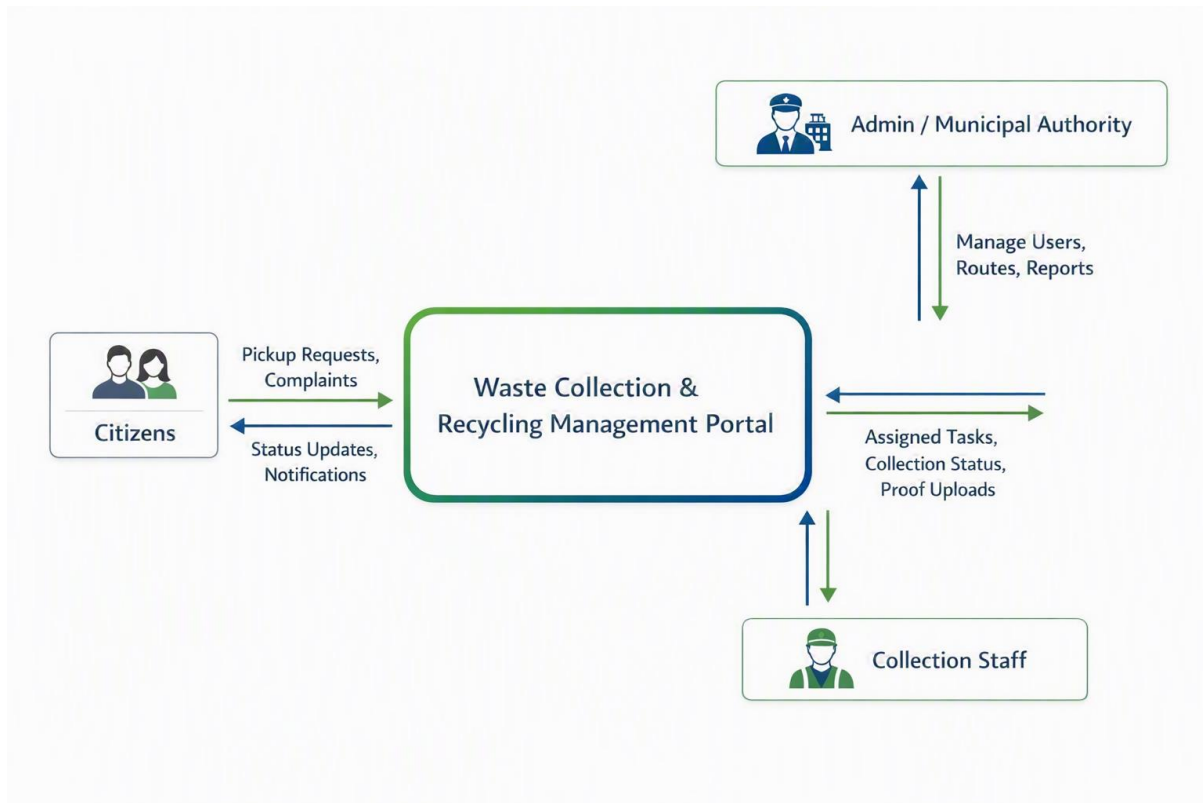
- View assigned routes and schedules
- Update collection status
- Report issues (blocked roads, overfilled bins)
- Upload proof (images, notes)

Administrators (Municipality / Management)

- Manage users and roles

- Assign routes and schedules
 - Monitor real-time operations
 - Track complaints and resolutions
 - Generate analytical reports
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System Context Diagram (High-Level View)



External Entities:

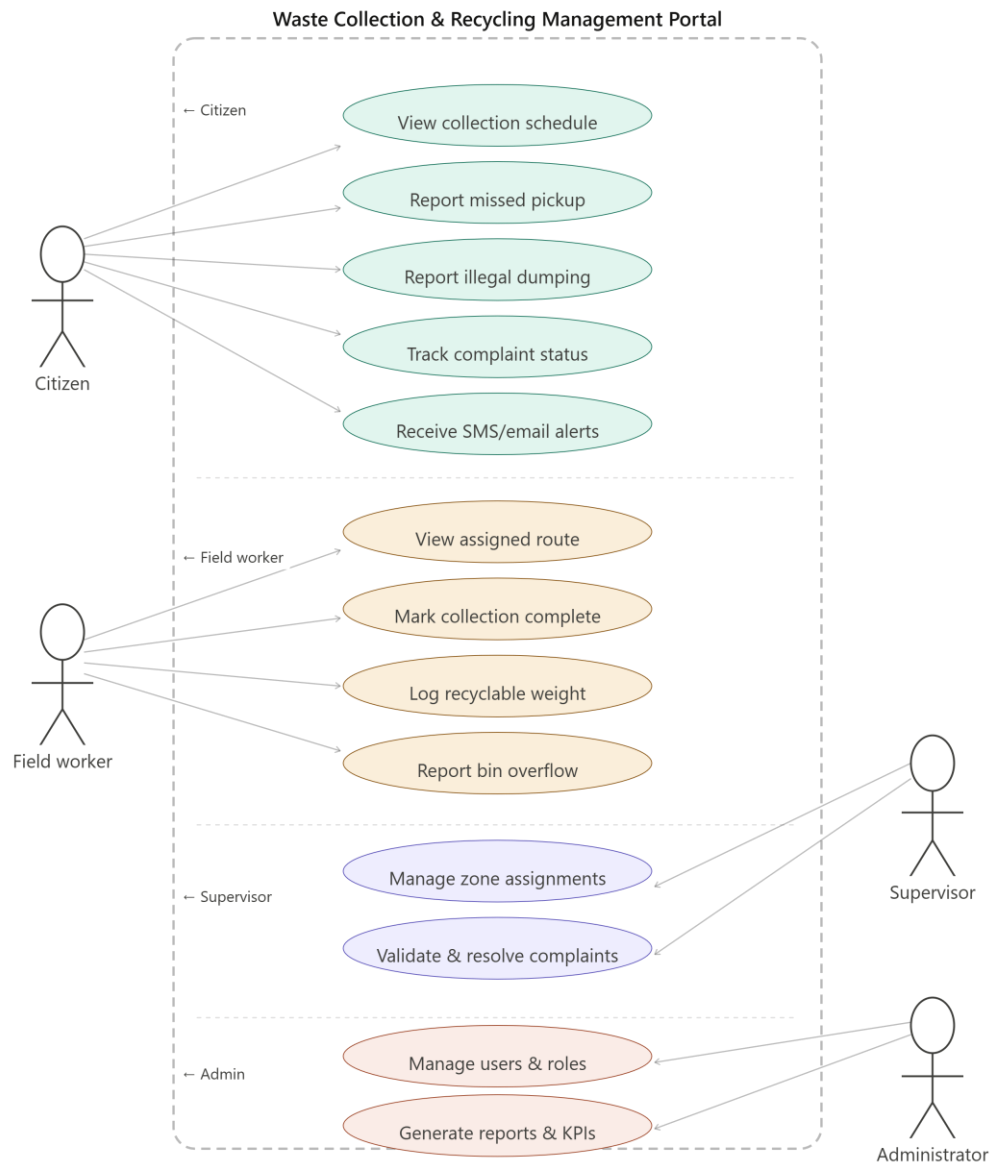
- Citizens
- Collection Staff
- Admin/Municipal Authority

System: Waste Collection & Recycling Management Portal

Interactions:

- Citizens → Submit requests, complaints
 - Staff → Update collection status
 - Admin → Manage, monitor, analyze
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Use Case Diagram (Functional Scope)



Citizen Use Cases

- Register / Login
- Submit Pickup Request
- Report Complaint
- Track Status
- Receive Notifications

Staff Use Cases

- Login
- View Assigned Tasks
- Update Collection Status

- Upload Proof

Admin Use Cases

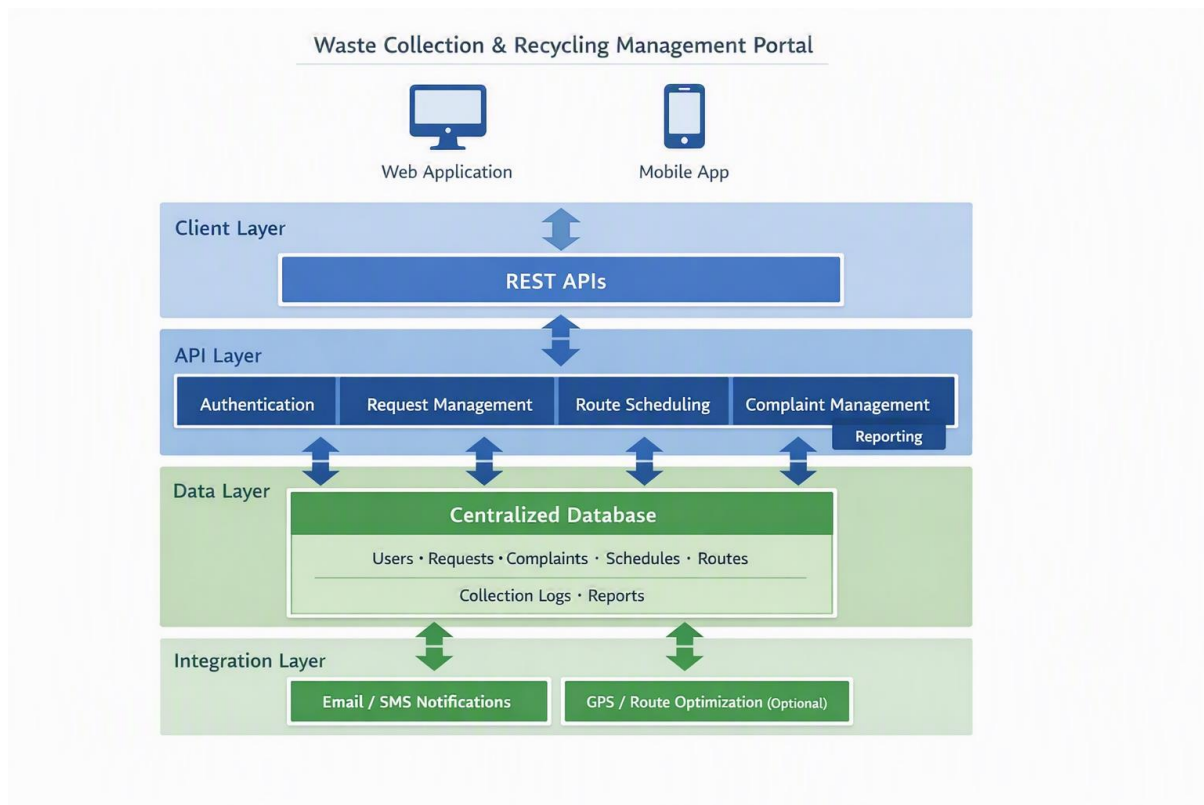
- Manage Users
 - Assign Routes
 - Monitor Collection
 - View Reports
 - Resolve Complaints
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Functional Flow Diagram (Waste Lifecycle)



1. Citizen submits waste pickup request / complaint
 2. System logs and categorizes request
 3. Admin assigns task to collection staff
 4. Collection staff performs pickup
 5. Staff updates status (Completed / Pending)
 6. System notifies citizen
 7. Data stored for reporting & analysis
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System Architecture Diagram



Frontend

- Web Application (React / Angular)
- Mobile App (optional)

Backend

- Application Server developed using Java
- REST APIs

The backend logic, business rules, and REST API services will be implemented using Java to ensure reliability, scalability, and maintainability.

Database

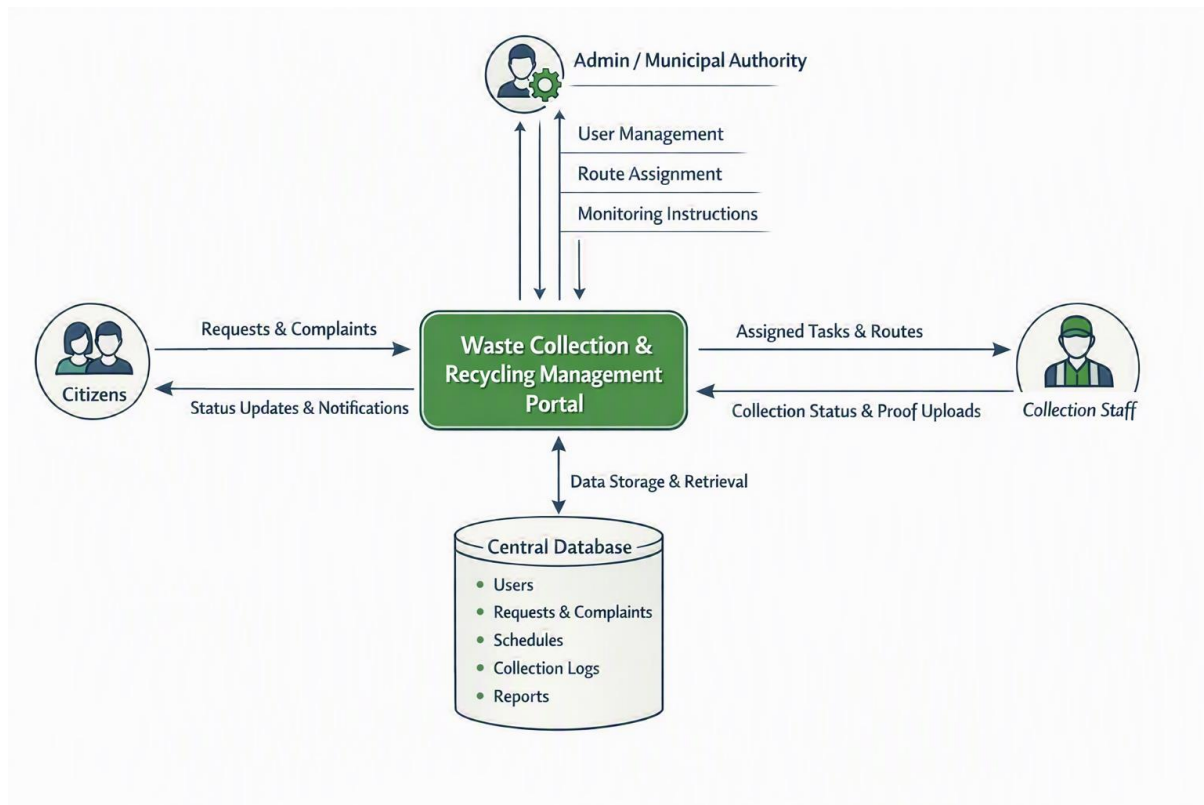
- Centralized Database using MySQL
- Stores users, waste requests, collection logs, reports

MySQL will be used to store and manage user records, waste pickup requests, complaints, collection schedules, route details, status updates, and generated reports.

Integration Components

- Notification System (Email/SMS)
- GPS/Route Optimization (optional)

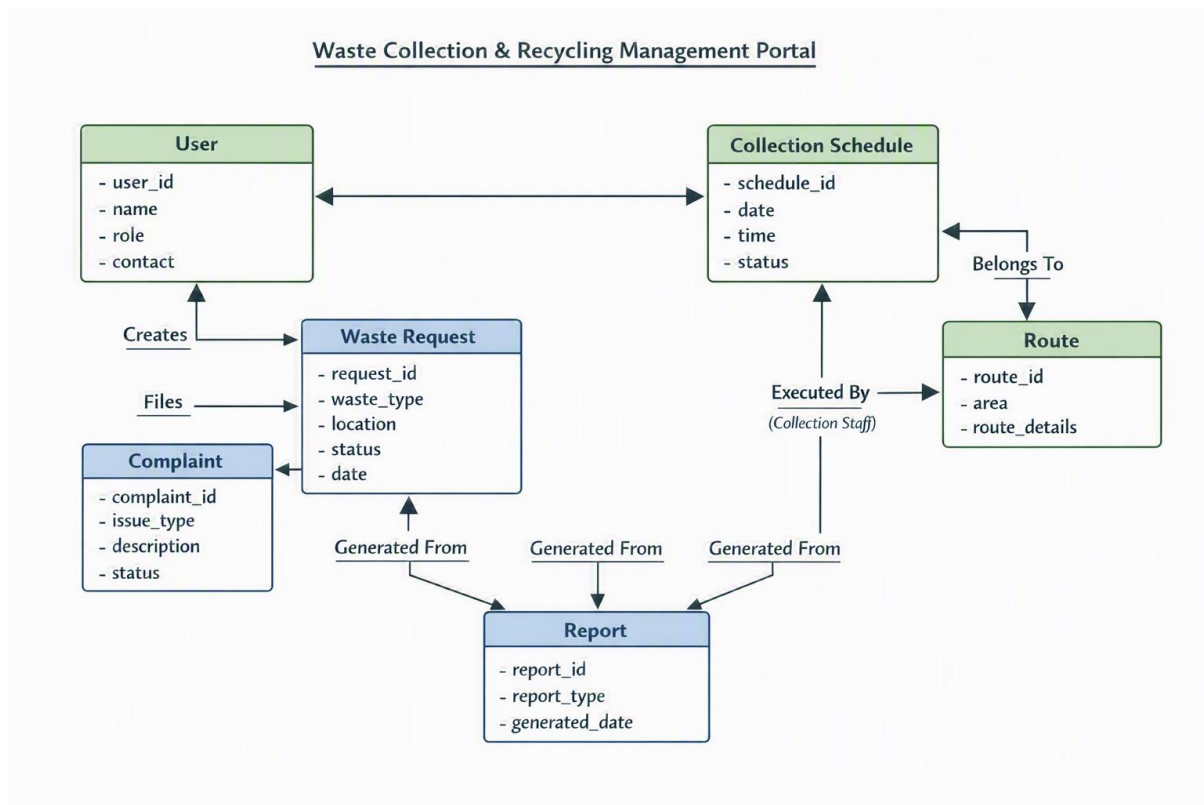
DFD (Data Flow Diagram)



Level 0 (Context Level)

- Citizens → Requests/Complaints → System
 - System → Updates/Notifications → Citizens
 - Admin ↔ System (management & monitoring)
 - Staff ↔ System (task execution updates)
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ER Diagram (Entities & Relationships)



Entities

- User (Citizen, Staff, Admin)
- Waste Request
- Complaint
- Collection Schedule
- Route
- Report

Relationships

- User creates Waste Request
- Admin assigns Schedule to Staff
- Staff executes Collection
- System generates Reports

Conclusion

The **Waste Collection & Recycling Management Portal** will modernize waste management operations by:

- Improving efficiency

- Ending transparency
- Promoting recycling awareness
- Enabling data-driven decision making

This system will ultimately contribute to cleaner cities and sustainable waste management practices.
